arXiv:2105.08328v1 [cs.RO] 18 May 2021

基于仿真到现实强化学习的盲态双足机器人楼梯穿越

Jonah Siekmann*†, Kevin Green*, John Warila*, Alan Fern*, Jonathan Hurst* †
*CoRIS研究所，俄勒冈州立大学{greenkev,
warilaj,afern,jhurst}@oregonstate.edu
† 敏捷机器人{jonah.siekmann}
@agilityrobotics.com

摘要——准确而精细的地形估计是机器人在真实环境中运动的一大难题。因此，设计不依赖精确估计、以免变得脆弱的系统十分有用。本文通过研究双足机器人在没有任何外部感知或地形模型的情况下穿越楼梯类地形的问题，探索这类方法的极限。对这种盲态双足平台来说，由于高度变化难以预料，问题显得很困难（即使对人类也一样）。我们的主要贡献是证明：仅利用本体感觉反馈，***sim-to-real***（仿真到现实）强化学习（***RL***）就能让双足机器人 ***Cassie*** 在楼梯类地形上实现鲁棒运动。重要的是，这只需对现有的平地训练 ***RL*** 框架进行修改，加入楼梯类地形随机化，而无需改变奖励函数。据我们所知，这是首个仅依靠本体感觉即可可靠穿越多种真实楼梯及其他楼梯类扰动的双足、人体尺度机器人控制器。

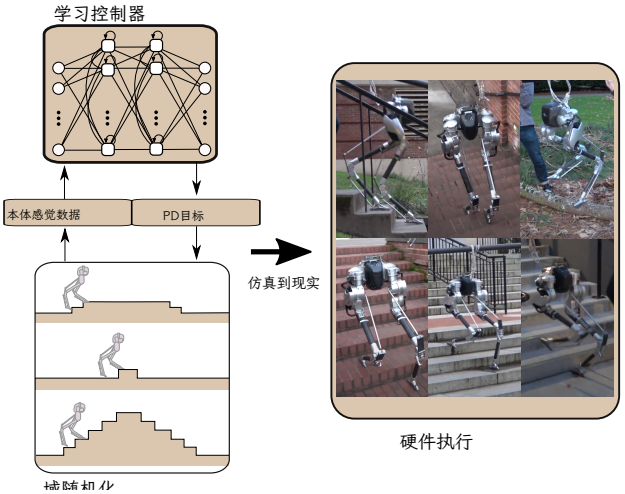


图1：在这项工作中，我们探究了盲态双足运动的极限。我们提出了一种训练流程，能够生成在现实世界中盲态上下楼梯的策略。这些策略学习本体感觉反射，以抵抗地面高度的显著扰动，从而在众多真实世界环境中表现出高度鲁棒的行为。

I. 引言

为了在现实世界中发挥作用，双足机器人和人形机器人需要能够上下楼梯及类似楼梯的地形，例如凸起的平台或突然的垂直落差，这些都是以人为中心的环境中的常见特征。能否稳健地穿越这些环境，对于让机器人安全地与人类协同工作至关重要。然而，要在双足平台上实现这种稳健性并非易事；虽然四足机器人等其他平台凭借同一时刻与地面的多个接触点以及像桌子一样能够停下站立的特性而具有先天稳定性，但像 Cassie 这样的双足机器人则完全依赖动态稳定性（本质上始终处于将要摔倒的状态）。在类似楼梯的环境中，这一点尤其明显，因为仅凭两条腿很难从失误中恢复过来。

相比之下，具有四足形态的机器人已经能够仅凭本体感觉来通过楼梯 [1, 2],
此外，六足机器人甚至已经能够通过开环控制上下楼梯 [3]。虽然平面双足机器人已被证明能够抵抗诸如较大的意外下台阶等扰动 [4], , 但旨在让此类机器人在现实世界中穿越楼梯的绝大多数方法要么需要精确的视觉系统 [5, 6, 7] ，要么需要在严格受控的实验室环境中运行[8, 9, 10], , 这意味着该机器人的定位依赖于一个已知的起始位置，或者楼梯是配合机器人形态设计的。

然而，机器人必须能够在受控的实验室条件之外运行，并应对现实世界中千变万化的环境。这一目标与完全依赖RGB相机和深度相机等外部传感器来准确估计地形的做法并不相容，因为这种依赖会在现实世界条件下引入脆弱性 [11]。例如，相机在遇到遮挡、雾气或变化的光照条件时可能不可靠。此外，将最先进的计算机视觉系统集成到高速控制器中在技术上也很困难，尤其是在移动机器人这类计算能力受限的平台上。出于实际考虑，底层控制器应在尽可能少依赖外界信息的前提下尽量鲁棒。理想情况下，双足机器人应尽可能依靠本体感觉穿越人类环境中尽可能多的区域，同时依靠外部感知来提高效率并进行高层规划（并能够承受错误感知的影响）。这就引出了一个问题：能有多鲁棒，

一只无视觉双足机器人？

基于强化学习（RL）的方法已开始在鲁棒的现实世界腿式运动方面展现出显著前景 [1, 12, 13]。与依赖预设地面接触时间表或基于力的事件检测的优化或启发式控制方法不同，RL 能够生成控制策略，这些策略通过训练中接触各种扰动，学习本体感觉反射以及应对意外过早或过迟接触和粗糙地形的策略。然而，这种方法的局限性尚不明确，且先前的工作尚未在与楼梯类地形相关的扰动规模和多样性上得到验证。

在这项工作中，我们展示了针对复杂楼梯类地形的鲁棒本体感觉双足控制可以通过现有的强化学习框架以极小且出人意料的修改得以学会。具体而言，唯一需要的调整是训练中使用的地形随机化，其中我们定义了包含上行和下行楼梯的分布，涵盖接触平面的高度、宽度和坡度变化。基于该分布的学习使机器人能够在未知楼梯上实现盲式上下行走，并处理更具一般性的楼梯类地形特征，例如原木、路缘石、落差等。所学的控制器已在仿真和多种真实世界环境中得到验证。据我们所知，这是此类研究的首次展示，也表明应继续探索鲁棒本体感觉双足控制的边界。

II. 强化学习问题表述

我们采用一种仿真到现实的强化学习（RL）方法来学习双足运动，并假定读者对 RL [14] 已有基本了解。在一般 RL 设定中，在每个离散时间步 t，机器人控制策略 π 接收当前状态 s_t 并返回动作 a_t；该动作被执行后使系统转移到下一个状态 s_{t+1}。状态转移的动态特性对机器人是未知的，它由地形类型等环境条件以及机器人自身的动力学特性共同决定。此外，在学习过程中，每次状态转移都会关联一个实值奖励 r_t。该奖励由应用目标决定，以鼓励学习过程中的期望行为。本文考虑的 RL 优化目标是：通过与环境的交互学习一个策略，使其在有限时域 T 上最大化期望累积折扣奖励。即寻找一个策略 π 以最大化：$J(\pi) = \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R_t\right]$，其中 $\gamma \in [0,1]$ 是折扣因子，R_t 是一个随机变量，表示在从初始状态分布中抽取一个状态后遵循 π 在时间 t 获得的奖励。

对于复杂环境，强化学习通常需要大量训练经验才能找到好的策略。此外，对于双足运动，训练过程会涉及许多跌倒和碰撞，尤其是在训练早期。因此，在现实世界中从零开始训练并不可行，我们转而采用 sim-to-real（仿真到现实）强化学习范式。训练完全在仿真环境中完成，并使用动力学随机化（见下文），随后将所得策略用于现实世界。

在本节其余部分，我们将详细介绍本文所采用的特定 sim-to-real 强化学习方案，其基础是近期关于在平坦地形上学习不同双足步态的研究工作 [13] 。令人惊讶的是，要让策略学习适用于本文中复杂得多的类楼梯地形[1] ，只需进行极小的改动。特别是，唯一需要的主要修改是随机生成楼梯状而非多为平坦的地形域，如后文第 III 节所述；此外无需任何新的楼梯专用奖励项。

A. 状态空间

每个时间步输入给控制策略的状态 s_t 包含三个主要部分。首先，该状态包含机器人瞬时的物理状态信息，包括以四元数表示的骨盆姿态、骨盆角速度、关节位置和关节速度。s_t 的第二部分由来自人类操作员的指令输入构成。这些指令在训练期间会被随机化，以让策略获得在各种速度和接近角度下尝试穿越楼梯的广泛经验。该随机化的详细信息见表 I。

指令	变化概率	范围
前进速度	1/300	[-0.3m/s，1.5m/s]
横向速度	1/300	[-0.3米/秒，0.3米/秒
转向速率	1/300	[-90deg/s，90deg/s]

表 I：在每个时间步，输入到策略的每个命令都有 1/300 的概率被改变。当这种情况发生时，会从由最右侧列参数化的均匀分布中采样一个新命令。鉴于最大回合长度为 300 个离散时间步，这意味着每个命令平均每个回合至少会改变一次。

第三部分包含两个循环时钟输入，每个输入分别对应机器人的一条腿，p：

$$p = \begin{cases} \sin(2\pi(\phi_t + 0.0)) \\ \sin(2\pi(\phi_t + 0.5)) \end{cases} \quad (1)$$

这里 ϕ_t 是一个相位变量，它从 0 递增到 1，然后循环回 0，用于跟踪步态的当前相位。常量偏移量 0.0 和 0.5 是相位偏移，用于确保运动过程中左右腿在相位上始终保持完全相反（即相差半个周期）。

B. 动作空间

控制策略在每个时间步（运行频率为 40Hz）输出的动作 a_t 是一个 11 维向量，其中前 10 个分量对应各关节的 PD 目标，每个目标都会被输入到以 2KHz 运行的对应关节 PD 控制器中。已有研究发现，在 PD 目标空间中进行动作学习，比直接学习更高频率的执行指令 [15] 更为有利。

[1]这一点是在我们首次在类似楼梯的地形上取得成功后，经过仔细的消融分析才发现的；而最初的成功包含了看似必要的对先前工作的修改，例如更复杂的奖励函数和状态特征。

$_t$ 的最后一个维度是时钟增量 δ_t（有关时钟的信息请参阅 II-A），它允许策略调节步态的迈步频率。直观地说，这使控制器能够针对特定的步态、指令和地形选择合适的迈步频率。具体而言，状态表示（第 II-A 节）中的相位变量 ϕ 在每个时间步 t 按以下方式更新：

$$\phi_{t+1} = \text{fmod}(\phi_t + \delta_t, 1.0). \quad (2)$$

该增量以一种受限的方式界定，使得策略可以在标称步频的0.5倍到1.5倍之间调节步态周期（标称步频约为每0.7秒一个步态周期）。虽然该分量包含在控制策略的动作中，但它似乎对性能没有显著影响，并且学习到的策略在响应扰动时不会在 δ_t 上产生很大变化。我们推测，未来的消融分析将表明它对真实机器人上的性能并不重要。[2]

C. 奖励函数

我们使用 [13] 中介绍的方法来设定奖励函数。简要回顾该方法：我们希望得到一种奖励框架，使策略在特定时间内因环境中某些量的幅值较大而受到惩罚，同时允许这些量在其他时间保持较大值。我们将足底力和足部速度指定为这样两个量；惩罚足底力会激励策略抬脚，而惩罚足部速度会激励策略落脚。在这些基础奖励项之上，我们增加了额外的代价项，其中包括激励策略匹配平动速度和朝向的代价。我们还采用了鼓励动作平滑、提高能量效率以及减少骨盆晃动的代价。关于所用奖励函数的详细说明，请参见附录。与 [13]，一样，我们不依赖专家参考轨迹来学习行为。

D. 动力学随机化

为了克服模拟 Cassie 环境中可能存在的建模误差，我们沿用先前工作 [16] [13] 的做法，在每次训练回合开始时对若干重要的动力学参数进行随机化。这些随机化参数列于表 II。

E. 策略表示与学习

我们将控制策略表示为一个LSTM循环神经网络 [17]，，其中包含两个维度为128的循环隐藏层。我们之所以选择使用具备记忆能力的网络，是因为先前的研究表明，这类网络在处理部分可观测环境时具有更高的熟练度 [18] [16] [19]。对于涉及非基于记忆的控制策略的消融实验，我们使用一个标准的前馈神经网络，包含两个维度为300的隐藏层，并使用tanh激活函数，

[2]我们在此先将其作为一个假设，因为在提交时我们尚未能在真实机器人上进行测试。

参数	Unit	范围
关节阻尼	Nms/rad	$[0.5, 3.5]$ ×默认值
关节质量	kg	$[0.5, 1.7]$ ×默认值
地面摩擦力	–	$[0.5, 1.1]$
关节编码器偏移	rad	$[-0.05, 0.05]$
执行频率	Hz	$[37, 42]$

表 II：为了防止对仿真动力学过拟合，并促进平滑的仿真到现实迁移，我们采用了动力学随机化。上述范围分别为每个所列参数定义了均匀分布。阻尼、质量、摩擦力和编码器偏移在每次 rollout 开始时随机化，而执行频率在每个时间步随机化，以模拟真实机器人上可变系统延迟的影响。

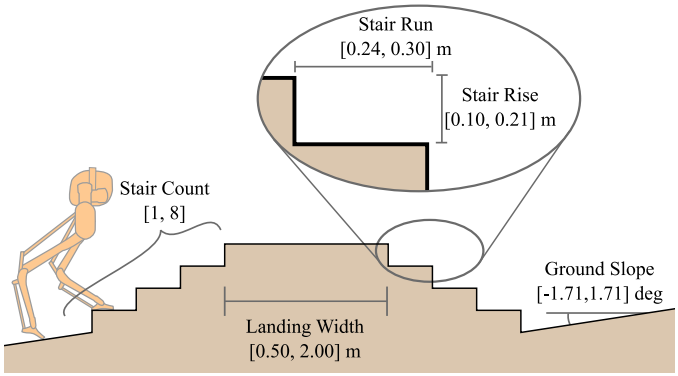


图 2：为了确保对多种可能的类楼梯地形具有鲁棒性，我们在仿真中每个回合开始生成楼梯时随机化若干参数。这些参数包括楼梯级数、每级楼梯的高度、每级楼梯的长度、楼梯顶部平台的长度，以及楼梯前后紧邻地面的坡度。

从而使其参数数量约等于LSTM网络的参数数量。

在策略的仿真到现实训练中，我们使用近端策略优化（PPO） [20]，，一种无模型深度强化学习算法。具体地，我们采用一种以 KL 散度阈值作为终止条件的变体：每次更新策略时，计算更新后策略与更新前策略之间的 KL 散度，若散度过大则中止本次更新。在训练过程中，我们使用镜像损失项 [21] ，以确保控制策略不会学习到不对称步态。对于循环策略，我们从经验回放缓冲区中按回合采样批次，如 [19], 中所述；对于前馈策略，我们则按时间步采样批次。每个回合限制为 300 个时间步，约对应 7.5 秒的仿真时间。

III. 地形随机化

先前将强化学习应用于Cassie的工作要么在平坦地面上训练 [12] [19] ，要么在随机轻微斜坡上训练 [13]。其他深度强化学习工作研究了采用难度随训练进展而逐渐增加的地形课程 [1]。为了简化起见，我们发现无需课程，直接在随机楼梯的交互上进行训练，就足以学习到稳健的行为。

为此，我们在一个平面上进行训练，该平面在每次rollout开始时沿俯仰轴和翻滚轴随机化倾斜角度，范围在-0.03弧度到0.03弧度之间。作为动力学随机化的一部分，地面摩擦也会被随机化，从而增加环境的潜在难度。楼梯的起始位置在每次rollout开始时随机化，因此回合可以从策略已经位于楼梯顶部开始，也可以从楼梯位于策略前方最多10米处开始。这样做是为了确保策略能够在平坦或倾斜地面以及楼梯上获得大量经验。

楼梯的尺寸在典型城市建筑规范的尺寸范围内随机化：每级台阶的踏步高度在10cm至21cm之间，踏步宽度在24cm至30cm之间。楼梯的台阶数也是随机化的，因此每段楼梯包含1到8级台阶。每一步的踏步高度和踏步宽度都会增加少量噪声（± 1cm），使楼梯永远不会完全一致，从而避免策略通过本体感觉推断出楼梯的精确尺寸，并因此对完全一致的楼梯产生过拟合。

IV. 结果

我们训练了四组策略，每组由五个使用不同随机种子初始化的策略组成。首先，我们训练了一组采用楼梯地形随机化的简单LSTM策略，本节中称为Stair LSTM。为了研究记忆的重要性，我们还训练了一组同样采用楼梯地形随机化的前馈策略，记为Stair FF。我们还训练了一组不采用楼梯地形随机化的策略，记为Flat Ground LSTM，以研究本文引入的地形随机化的重要性。最后一组策略在训练时使用了一个额外且简单的二值输入，该输入告知策略其周围一米范围内是否存在楼梯，这组策略称为Proximity LSTM，以研究将外部世界信息泄露给策略的益处。

每个策略都被训练至从虚拟环境中采样满3亿个时间步为止；该虚拟环境使用 MuJoCo[22] 进行仿真。我们为 PPO 算法选择的超参数包括：经验回放缓冲区大小为 50,000 个时间步；循环策略的批大小为 64 条轨迹；前馈策略的批大小为 1024 个时间步。每个经验回放缓冲区最多采样五个轮次，若 KL 散度达到最大允许阈值 0.02，则提前终止优化。每次迭代开始时，我们都会清空经验回放缓冲区。我们使用 Adam [23] 优化器，actor 和 critic 的学习率均为 0.0005；二者分别训练，不共享参数。

A. 仿真

1) 成功上下楼梯的概率：为了理解记忆和地形随机化的重要性，我们在成功攀爬和下行一组楼梯的任务上评估了三组策略。

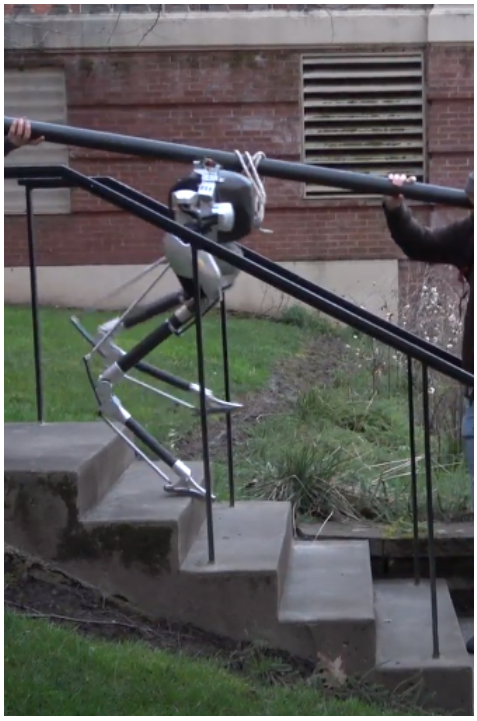


图 3：习得策略对多种类楼梯地形表现出高度的盲鲁棒性，并且能够可靠地上下人类环境中常见尺寸的楼梯。

在仿真中。我们比较了 Stair FF、Stair LSTM 和 Flat Ground LSTM 策略在此任务上的表现。

具体来说，我们执行了 150 次试验，测试策略能够成功爬上一段五级台阶的频率；该段楼梯的每级台阶高 17cm、深 30cm（这是一种现实世界中典型且相对平缓的楼梯几何形状）。这应能让我们估计每组控制策略在盲态接近一段楼梯时，能够多可靠地爬上楼梯。成功定义为不摔倒并到达楼梯顶端。下楼梯也采用同样的流程：我们在尺寸相同的楼梯上执行 150 次试验，并记录每组策略在不摔倒的情况下到达底部的成功率。

三种不同训练条件下的测试结果如图4所示。我们注意到，Stair LSTM 策略的总体成功率最高。尽管如此，成功率在很大程度上取决于接近速度。这些策略在低速时失败率较高，因为此时可能缺乏足够动量来越过选择不当的落脚点。在高速时失败率也较高，这可能是由于高速步态更具动态性。

Flat Ground LSTM 策略在训练中从未见过类楼梯地形，因此无法进行补偿，在上楼梯和下楼梯时都具有很高的失败率。Stair FF 策略尽管在训练中遇到过楼梯，却无法学会有效的楼梯应对策略，这说明记忆可能是对类楼梯地形具备鲁棒性的重要机制。

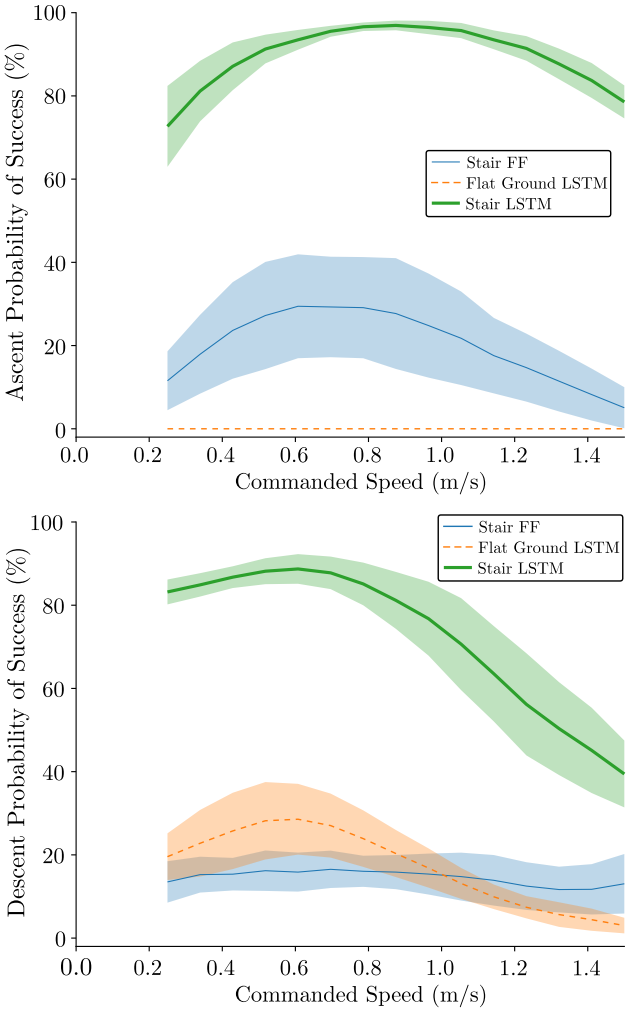


Fig. 4: We evaluate the probability of successfully climbing and descending stairs without falling as a function of commanded speed between 0.25 m/s and 1.5 m/s over 150 trials. For Stair LSTM policies, there seems to be an optimal approach speed for climbing stairs and a separate optimal descent speed. Stair FF policies do not attain high performance, implying that memory could be an important component of the learned behavior. Flat Ground LSTM policies, having never encountered stairs in training, are virtually unable to climb stairs while finding some success in descending stairs without falling over.

2) Energy Efficiency Comparison: To understand the consequences of training with terrain randomization, we also compare the cost of transport between Flat Ground LSTM policies, Stair LSTM policies, and Proximity LSTM policies. The cost of transport (CoT) is a common measure of efficiency of legged robots, humans and animals. It is the energy used per distance, normalized by weight to be unitless. It is defined as

$$\text{CoT} = \frac{E_\text{m}}{Mgd}, \tag{3}$$

where E_m is the energy used by the motors, M is the total mass of the robot, g is the gravitational acceleration and d is the distance traveled. The energy used by Cassie is calculated using positive actuator work and resistive losses via

$$E_\text{m} = \int_0^T \Big(\sum_i max(\tau_i \cdot \omega_i, 0) + \frac{\omega_i^\text{max}}{P_i^\text{max}} \tau_i^2 \Big) dt. \tag{4}$$

Here τ_i is the torque applied to motor i and ω_i is its rotational velocity. We use two parameters to define the resistive losses in terms of torque, P_i^{max} is the maximum input power and ω_i^{max} is the maximum speed of motor i. The results of testing steady state CoT at 1 m/s on flat ground can be seen in Table III. These calculations of CoT do not include the overhead power draw from computation and control electronics so they should not be used to compare between robots, only between control policies.

Policy Group	Mean CoT	Std. CoT
Proximity LSTM (stairs)	0.47	0.0086
Stair LSTM	0.46	0.0323
Proximity LSTM (flat)	0.39	0.0257
Flat Ground LSTM	0.38	0.0205

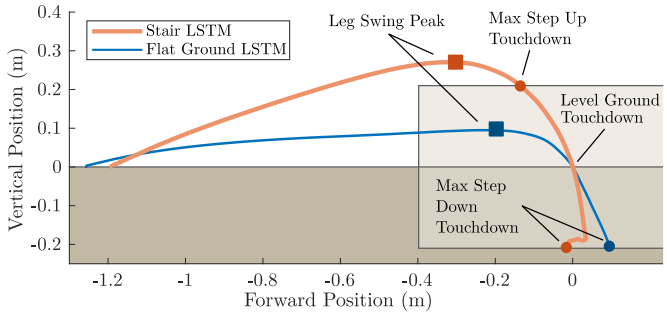
TABLE III: Locomotion efficiency as measured by cost of transport (CoT) for walking at 1 m/s over flat ground in simulation between three groups of policies over all five random seeds. We note that policies not trained on stair terrain randomization tend to learn more energy efficient gaits, though some energy efficiency can be recovered by providing the stair-trained policies with a binary stair presence/absence input.

We find that Flat Ground LSTM policies learn the most energy efficient gaits for walking on flat ground. Stair LSTM policies learn less efficient flat-ground gaits in order to be robust to stairs; however, the stair proximity input to the Proximity LSTM can help to recover some of this lost energy efficiency by allowing the learned controller to switch between a stair-ready gait and a more energy efficient, flat-ground gait.

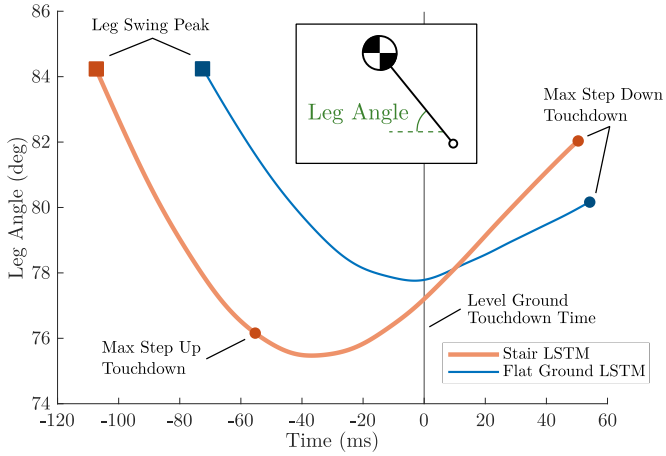
B. Behavior Analysis

To understand the strategy adopted by the policy, we can benefit from taking the perspective of experimental biology.We specifically look at the behavior as the robot contacts the first step up or down after walking along flat ground. First we will analyze the swing leg motion to understand how the robot places its foot on step ups and step downs. Once the swing foot contacts a step up or down, the force applied by the foot on the ground during stance phase can be modulated to better prepare the robot for future steps. We analyze how the ground reaction force and total impulse varies in the case of step ups and step downs.

1) Swing Foot Motion: To understand the change the stair terrain makes in the foot swing path we compare the result of a Flat Ground LSTM policy and a Stair LSTM policy when they encounter a drop step. The foot swing path during a drop step lets us see where the policy would place the foot if it had encountered a step up or a step down. Fig. 5a shows the foot swing path of these two policies relative to the ground. We can see that the Stair LSTM policy takes a much higher step compared to the Flat Ground LSTM policy which gives it additional clearance so it can step up onto a large step. A second interesting observation is the steeper path of the swing



(a) Swing foot paths for the stair trained policy and the flat ground policy overlaid on example step ups and step downs.



(b) The leg angle between the robot body and the swing foot as the foot descends toward touchdown.

Fig. 5: A comparison of the swing foot motion of the Stair LSTM policy and the Flat Ground LSTM policy while locomoting at 1.0 m/s. There is a significant change in the leg swing policy as a result of training on randomized stairs. The most significant changes are higher foot clearance, a steeper foot descent and a faster leg angle retraction rate.

foot for the Stair LSTM policy. The swing foot only moves forward 14 cm while it is in the height range where it may encounter the front face of a step up. We hypothesize this is a strategy that prevents the foot from stubbing the toe too hard on the front face of a stair and causing the robot to trip forward.

A second viewpoint to understand leg swing motion is to look at the leg swing retraction. In humans and in bipedal birds it is observed that the swing leg is swung backwards, relative to the body, towards the ground near the end of stance [24, 25]. This has the benefits of reducing the velocity of the foot relative to the ground and thus reducing the impact [26] as well as improving ground height disturbance rejection by automatically varying the leg touchdown angle [27].

Our training procedure does not explicitly incentivize the policy to exhibit these leg swing retraction behaviors, but we do see them emerge as shown in Fig. 5b. This figure shows the angle of the swing foot relative to the body between the peak of leg swing and contact with the maximum step down. The Stair LSTM policy has a faster leg retraction rate compared to the Flat Ground LSTM policy. With only this data we cannot say if this retraction profile is optimal or even if it is the cause of the improved performance on stairs. However, the fact that there is a significant change in the leg retraction profile as a result of training on stairs is an interesting observation.

2) Ground Reaction Forces: Once the robot's foot has touched down its control authority is limited due to the underactuated nature of bipedal locomotion. However, the robot still has a significant amount of control through the ground reaction force. To understand how the Stair LSTM policy reacts to a 10 cm step up or down we plot the horizontal and vertical ground reaction forces in the sagittal plane in Fig. 6. At the beginning of stance there is a large spike in force that dwarfs the normal forces during stance. The force value during this spike is largely defined by the tuning of the simulation contact model so it is not of primary interest to understanding the behavior of the policy. The first interesting thing we see in subplot A is that the maximum nominal leg force is held relatively constant which is a predicted result of a well adjusted leg swing policy [28]. Second we see that the magnitude of the second hump in the double humped ground reaction forces is increased in the step down and decreased in the step up. In the horizontal forces (subplot B) we see an oscillating signal where the oscillations match the frequency of policy evaluation. We hypothesize that this is the policy working to control the attitude of the pelvis. Prioritizing body attitude over forward velocity would be similar to the explicit priorities during single stance in Virtual Model Control [29]. The lower two subplots (C and D) show the cumulative impulse in the vertical and forward directions throughout the stance phase. We can see that the step up applies a larger vertical impulse and the step down a smaller vertical impulse. This agrees with the intuition that the robot should apply a smaller vertical impulse to lower itself down a step compared to lifting itself up a step. The horizontal impulse tells us if the robot speeds up or slows down in the forward direction during the stance phase. We see that the step down results in a significantly larger forward impulse and the step up reduces the vertical impulse very slightly. This aligns with the predicted behavior from a well tuned swing leg retraction policy.

C. Hardware

The recurrent policies transferred to hardware without any notable difficulties. We were able to take the robot for a walk around a large university campus using a randomly selected Stair LSTM policy and attempt to climb the staircases we came across. We observed robust and error-correcting behavior, as well as successful and repeatable stair ascents and descents. In addition, we noted robustness to uneven terrain, logs, and curbs, none of which were modeled in training. The policy was similarly robust to inclines and deformable terrain, demonstrated by a walk through a wet grass field and up a small hill. These experiments can be seen in our submission

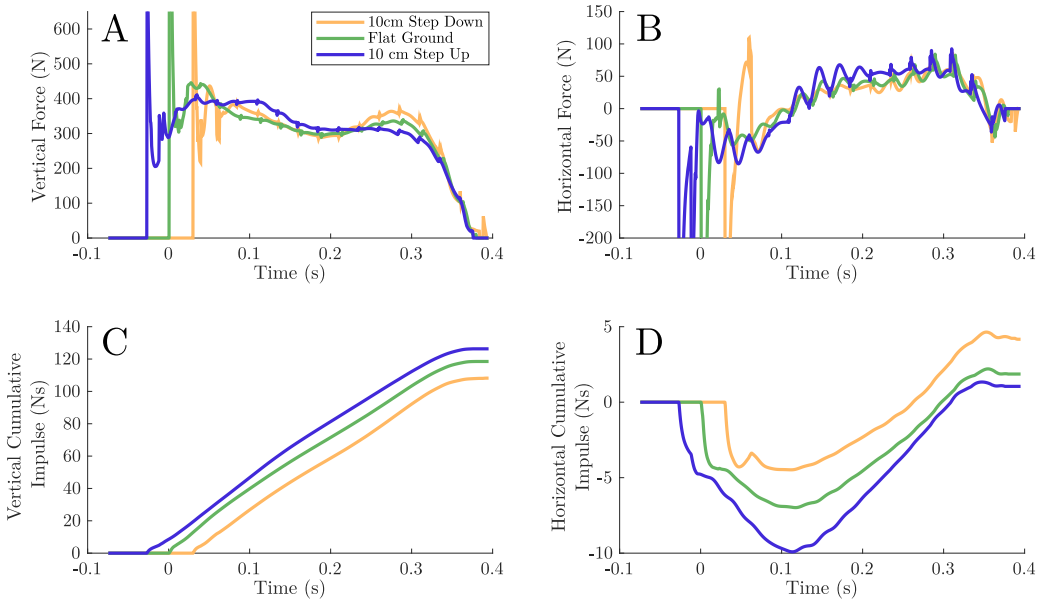


Fig. 6: The ground reaction forces and cumulative impulses of a Stair LSTM policy when it encounters varying ground height. The peak vertical force (A) after the impact remain roughly equivalent while the force in the second half of stance is modulated. The horizontal force (B) shows oscillations that match the frequency of the learned policy execution rate. This may be the policy controlling the body's attitude. The total vertical impulse (C) shows the expected result of a larger impulse stepping up and a smaller one stepping down. The horizontal impulse (D) shows a result that is predicted by leg swing retraction. When stepping down the foot is shifted backwards relative to the body which results in net acceleration forward which is shown here by a positive horizontal impulse.

video [3], and a still image of one such experiment can be seen in Fig. 3.

In addition to testing one-off terrains all over the university campus, we ran ten trials ascending stairs, and ten trials descending stairs on an outdoor real-world staircase. We recorded an 80% success rate in ascending stairs using the selected Stair LSTM policy, and a 100% success rate in descending stairs. A full video of this trial can be seen in our attachment to this submission. We note that the learned behavior is robust to missteps, and can quickly recover from mistakes, though the policy is not completely infallible and will fall if it makes a particularly egregious error. This experiment can be seen in our supplemental video [4]. The blind, proprioceptive learned strategy appears to rely on a solid stair face; evaluating policies on slatted stairs in simulation resulted in a much higher failure rate, pointing to the limits of such an approach. Even when explicitly included in training, slatted stairs tended to trip up policies on ascent. By contrast, stairs with randomly inclined steps (e.g., ones where each step had a unique pitch and roll orientation) did not seem to be difficult for ascent or descent. Likewise, approaching and ascending stairs at an angle did not seem to be an issue for policies.

V. Conclusion

In this work, we have motivated the desirability of a highly robust but blind walking controller, and demonstrated that such a blind bipedal walking controller is capable of climbing a wide variety of real-world stairs. In addition, we note that producing such a controller requires very little modification to an existing training pipeline [13], and in particular no stair-specific reward terms; simply adding stairs to the environment with no further information is sufficient for learning stair-capable control policies. An important requirement of this learned ability appears to be a memory mechanism of some kind, probably due to the partially observable nature of the task of walking through unknown terrain while blind. In future work, it will be interesting to investigate how vision can be most effectively used to improve the efficiency and/or performance of a blind bipedal robot. Further, this work has demonstrated surprising capabilities for blind locomotion and leaves open the question of where the limits lie.

[3][Web link to submission video: youtu.be/MPhEmC6b6XU]

[4][Web link to supplemental video: youtu.be/nuhHiKEtaZQ]

Acknowledgments

This work was partially supported by NSF grant IIS-1849343 and DARPA contract W911NF-16-1-0002. Special thanks to Intel for access to the vLab cluster, and to Yesh Godse, Jeremy Dao, and Helei Duan for advice and guidance.

Appendix

Reward Function

To briefly review the approach taken in [13], we wish to take advantage of the complementary nature of foot forces and foot velocities during locomotion to construct a reward function which will punish one and allow the other, and vice versa, at key intervals during the gait. We use a probabilistic framework to represent uncertainty around the timings of these intervals. More specifically, we make use of a binary-valued random indicator function $I_i(\phi)$ for each quantity q_i which

we wish to penalize at some time during the gait cycle. This indicator function is likely to be 1 during the interval in which it is active, and likely to be 0 during intervals in which it is not active. The distribution of this binary-valued random function is defined via the Von Mises distribution; for a more comprehensive description, see [19]. In addition, rather than use the actual random variable in the reward we instead opt to use its expectation for more stable learning; see Fig. 7 for a plot of this expectation.

Our full reward function is as follows;

$$R(s,\phi) = 1 - \mathbb{E}[\rho(s,\phi)] \tag{5}$$

Which is to say, our reward is the difference of a bias and the expectation of a probabilistic penalty term $\rho(s,\phi)$ as described in [13]. See Table IV for detailed information on the exact quantities and weightings used.

Weight	Cost Component
0.140	$1 - \mathbb{E}[I_{\text{left force}}(\phi)] \cdot \exp(-.01\|F_l\|)$
0.140	$1 - \mathbb{E}[I_{\text{right force}}(\phi)] \cdot \exp(-.01\|F_r\|)$
0.140	$1 - \mathbb{E}[I_{\text{left velocity}}(\phi)] \cdot \exp(-\|v_l\|)$
0.140	$1 - \mathbb{E}[I_{\text{right velocity}}(\phi)] \cdot \exp(-\|v_r\|)$
0.140	$1 - \exp(-\varepsilon_o)$
0.140	$1 - \exp(-\lvert\dot{x}_{\text{desired}} - \dot{x}_{\text{actual}}\rvert)$
0.078	$1 - \exp(-\lvert\dot{y}_{\text{desired}} - \dot{y}_{\text{actual}}\rvert)$
0.028	$1 - \exp(-5 \cdot \|a_t - a_{t-1}\|)$
0.028	$1 - \exp(-0.05 \cdot \|\tau\|)$
0.028	$1 - \exp(-0.1(\|\text{pelvis}_{\text{rot}}\| + \|\text{pelvis}_{\text{acc}}\|))$

TABLE IV: The cost terms which are summed together to compose the expected penalty, $\mathbb{E}[\rho(s,\phi)]$. Terms involving an expectation of a variable $I_i(\phi)$ vary over the course of the gait cycle, with the goal of penalizing foot forces and foot velocities at key intervals to teach the policy to lift and place the feet periodically in order to walk. Other terms exist for the sake of commanding the policy to move forward, backwards, or sideways, or turn the robot to face a desired heading. Finally, the remaining terms exist to reduce behaviors which are shaky and thus unlikely to work well on hardware.

We define F_l and F_r as the vectors of translational forces applied to the left and right foot, and v_l and v_r similarly as the vectors of left and right foot velocities. To maintain a steady orientation, an orientation error ε_o is used, which is equal to,

$$\varepsilon_o = 3(1 - \hat{q}^T q_{\text{body}})^2 + 10\left((1 - \hat{q}^T q_{\text{l}})^2 + (1 - \hat{q}^T q_{\text{r}})^2\right) \tag{6}$$

where q_{l} and q_{r} are the quaternion orientations of the left and right foot, q_{body} is the quaternion orientation of the pelvis, and $\hat{q}$ is a desired orientation (for our purposes, fixed to be always be facing straight ahead).

The quantities $\dot{x}_{\text{desired}}$ and $\dot{y}_{\text{desired}}$ correspond to a commanded translational speed, while $\dot{x}_{\text{actual}}$ and $\dot{y}_{\text{actual}}$ are the actual translational speed of the robot. The term $\text{pelvis}_{\text{rot}}$ represents the angular velocity while $\text{pelvis}_{\text{acc}}$ represents translational acceleration; these terms are used in the cost component to reduce the shakiness of locomotion behavior. The terms a_t and a_{t-1} refer to the current timestep's action and the previous timestep's action, and their use in the cost component is to encourage smooth behaviors. The term τ is the vector of net torques applied to the joints, and its use in the cost component is intended to encourage energy efficient gaits.

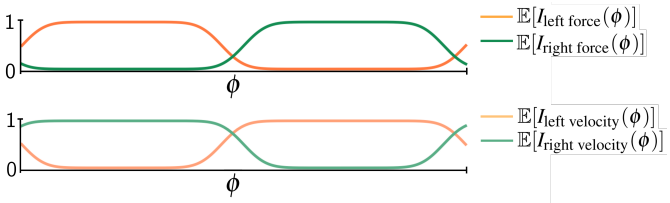


Fig. 7: By alternatingly punishing foot forces during a 'stance' phase to teach the policy to lift the foot, and punishing foot velocities during a 'swing' phase to teach the policy to place the foot on the ground, we can construct a foundation on which to learn simple walking behavior. Following in the path of previous work, we define these cyclic coefficents as random indicator functions of the phase, and take their expectation.

References

[1] Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning quadrupedal locomotion over challenging terrain. *Science Robotics*, 5(47):eabc5986, Oct 2020. ISSN 2470-9476. doi: 10.1126/scirobotics.abc5986.

[2] Gerardo Bledt, Matthew J Powell, Benjamin Katz, Jared Di Carlo, Patrick M Wensing, and Sangbae Kim. MIT Cheetah 3: Design and control of a robust, dynamic quadruped robot. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2245–2252. IEEE, 2018.

[3] EZ Moore and M Buehler. Stable stair climbing in a simple hexapod robot. Technical report, McGill Research Centre for Intelligent Machines, 2001.

[4] Hae-Won Park, Alireza Ramezani, and Jessy W Grizzle. A finite-state machine for accommodating unexpected large ground-height variations in bipedal robot walking. *IEEE Transactions on Robotics*, 29(2):331–345, 2012.

[5] J-S Gutmann, Masaki Fukuchi, and Masahiro Fujita. Stair climbing for humanoid robots using stereo vision. In *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)(IEEE Cat. No. 04CH37566)*, volume 2, pages 1407–1413. IEEE, 2004.

[6] Amos Albert, Michael Suppa, and Wilfried Gerth. Detection of stair dimensions for the path planning of a bipedal robot. In *2001 IEEE/ASME International Conference on Advanced Intelligent Mechatronics. Proceedings (Cat. No. 01TH8556)*, volume 2, pages 1291–1296. IEEE, 2001.

[7] Philipp Michel, Joel Chestnutt, Satoshi Kagami, Koichi Nishiwaki, James Kuffner, and Takeo Kanade. GPU-accelerated real-time 3D tracking for humanoid locomotion and stair climbing. In *2007 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 463–469. IEEE, 2007.

[8] Stéphane Caron, Abderrahmane Kheddar, and Olivier Tempier. Stair climbing stabilization of the HRP-4

humanoid robot using whole-body admittance control. In *2019 International Conference on Robotics and Automation (ICRA)*, pages 277–283. IEEE, 2019.

[9] Agility Robotics. Cassie: Dynamic Planning on Stairs.

[10] Giorgio Figliolini and Marco Ceccarelli. Climbing stairs with EP-WAR2 biped robot. In *Proceedings 2001 ICRA. IEEE International Conference on Robotics and Automation (Cat. No. 01CH37164)*, volume 4, pages 4116–4121. IEEE, 2001.

[11] Michele Focchi, Romeo Orsolino, Marco Camurri, Victor Barasuol, Carlos Mastalli, Darwin G. Caldwell, and Claudio Semini. *Heuristic Planning for Rough Terrain Locomotion in Presence of External Disturbances and Variable Perception Quality*, volume 132, pages 165–209. Springer International Publishing, Cham, 2020. ISBN 978-3-030-22327-4. doi: 10.1007/978-3-030-22327-4_9.

[12] Zhaoming Xie, Glen Berseth, Patrick Clary, Jonathan Hurst, and Michiel van de Panne. Feedback control for cassie with deep reinforcement learning. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1241–1246. IEEE, 2018.

[13] Jonah Siekmann, Yesh Godse, Alan Fern, and Jonathan Hurst. Sim-to-Real Learning of All Common Bipedal Gaits via Periodic Reward Composition. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[14] Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.

[15] Xue Bin Peng and Michiel van de Panne. Learning Locomotion Skills Using DeepRL: Does the Choice of Action Space Matter? *CoRR*, abs/1611.01055, 2016.

[16] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel. Sim-to-Real Transfer of Robotic Control with Dynamics Randomization. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3803–3810, 2018. doi: 10.1109/ICRA.2018.8460528.

[17] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.

[18] Nicolas Heess, Jonathan J Hunt, Timothy P Lillicrap, and David Silver. Memory-based control with recurrent neural networks, 2015.

[19] Jonah Siekmann, Srikar Valluri, Jeremy Dao, Lorenzo Bermillo, Helei Duan, Alan Fern, and Jonathan Hurst. Learning memory-based control for human-scale bipedal locomotion. In *Proceedings of Robotics: Science and Systems*, July 2020.

[20] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal Policy Optimization Algorithms, 2017.

[21] Farzad Adbolhosseini, Hung Yu Ling, Zhaoming Xie, Xue Bin Peng, and Michiel van de Panne. On Learning Symmetric Locomotion. In *Proc. ACM SIGGRAPH Motion, Interaction, and Games (MIG 2019)*, 2019.

[22] Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033. IEEE, 2012.

[23] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

[24] KL Poggensee, MA Sharbafi, and A Seyfarth. Characterizing swing-leg retraction in human locomotion. In *Mobile Service Robotics*, pages 377–384. World Scientific, 2014.

[25] Monica A. Daley and Andrew A. Biewener. Running over rough terrain reveals limb control for intrinsic stability. *Proceedings of the National Academy of Sciences*, 103(42):15681–15686, 2006. ISSN 0027-8424. doi: 10.1073/pnas.0601473103.

[26] Yvonne Blum, Susanne W Lipfert, Juergen Rummel, and André Seyfarth. Swing leg control in human running. *Bioinspiration & biomimetics*, 5(2):026006, 2010.

[27] André Seyfarth, Hartmut Geyer, and Hugh Herr. Swing-leg retraction: a simple control model for stable running. *Journal of Experimental Biology*, 206(15):2547–2555, 2003.

[28] Aleksandra V. Birn-Jeffery, Christian M. Hubicki, Yvonne Blum, Daniel Renjewski, Jonathan W. Hurst, and Monica A. Daley. Don't break a leg: running birds from quail to ostrich prioritise leg safety and economy on uneven terrain. *Journal of Experimental Biology*, 217(21):3786–3796, 2014. ISSN 0022-0949. doi: 10.1242/jeb.102640.

[29] Jerry Pratt and Chee-Meng Chew and Ann Torres and Peter Dilworth and Gill Pratt. Virtual model control: An intuitive approach for bipedal locomotion. *The International Journal of Robotics Research*, 20(2):129–143, 2001. doi: 10.1177/02783640122067309.